

The Coordination Tax Is Paid in Fuel, Not Time: Light-Speed Delay in a Self-Replicating Probe Swarm

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Abstract—A swarm of self-replicating probes fills a stellar field star by star, each probe settling a star and launching copies. The standard exploration-timescale model grants every probe perfect, instantaneous knowledge of which stars are already settled, and names finite light-speed as its own future work. We add that light-speed limit and measure its cost, and the first finding is a caution: a model advanced in a timestep coarser than the interval between probe launches makes many probes decide from the same stale snapshot at once, and the resulting collisions appear to slow the fill of the field by tens of percent. They do not. Resolved so that each probe decides at its true arrival time, light-speed lag does not measurably delay the fill, and the gravitational-slingshot speed-up survives contact with finite information, reproducing the source model quantitatively at about 166 times powered flight. The real cost lies elsewhere. The field is always filled, with a negligible change in the number of probes built; what the delay changes is the number of wasted journeys the swarm flies to stars other probes have already claimed. That is a tax in fuel and effort, not time, and it is governed cleanly by the probe’s speed in units of the speed of light: a median of about 0.7 percent of journeys at slingshot speeds, rising monotonically to about 18 percent at the directed-energy speeds beamed-propulsion concepts target, positive in all 32 seeds of a paired ensemble and a roughly scale-independent fraction of effort across a four-fold range of field sizes. Gravitational slingshots self-limit well below the speeds where the tax bites, so the cost falls specifically on fast, driven probes; and because the kinetic cost of a wasted hop grows with speed, the tax is heavier still as a fraction of the energy budget. The coordination cost a perfect-information model hides is paid in fuel, not time, and it scales with how fast the swarm flies.

Index Terms—self-replication, interstellar exploration, light-speed delay, multi-agent coordination, swarm dynamics

I. INTRODUCTION

A self-replicating probe does not explore a galaxy alone. It settles a star, builds copies of itself from local material, and sends them on to further stars, so a single launch grows into a front that sweeps outward and fills the reachable field. That a machine can build a copy of itself at all is von Neumann’s result, and a probe of this kind saturating the Galaxy in a few million years is the sharpest form of the Fermi question: if interstellar settlement is possible it should already be complete, so the observed silence is a paradox [1], [2]. How long the filling takes is the exploration-timescale question, and it has a long modelling lineage. Early work framed each interstellar probe as an independent agent acting on priors fixed before launch [3], and treated settlement as a front: a density-dependent diffusion with a travelling-wave

solution [4], or a Monte Carlo migration filling the Galaxy in tens of millions of years [5]. Later simulations made the field concrete, sending probes and subprobes through a defined stellar volume to measure fill times directly [6], [7], and established the seeded Monte Carlo methodology that situates an exploration timescale within a distribution of realizations rather than a single number [8].

The sharpest treatment for self-replicators is the slingshot line. Forgan, Papadogiannakis and Kitching showed that a single probe stealing energy from the motion of the stars it passes, a gravitational slingshot, explores up to two orders of magnitude faster than a probe under its own power [9]. Nicholson and Forgan extended that manoeuvre to self-replicating probes and confirmed the speed-up survives replication, with the surprising result that aiming for the nearest star can beat aiming for the largest boost [10]. The probe speeds that make this regime interesting, from the powered cruise of $3 \times 10^{-5} c$ up toward the 0.1 to 0.2 c of directed-energy or high-velocity concepts [11], [12], are exactly where the next assumption starts to hurt.

That assumption, which Nicholson and Forgan name as a limitation and leave for future work, is that every probe is granted perfect, instantaneous knowledge of which stars are already settled. No probe ever flies to a star another has just claimed, because it always knows. In a real galaxy no such oracle exists: news that a star has been settled can travel no faster than light, so a probe choosing its next target acts on a picture of the field that is, for distant stars, years or centuries out of date. This is the elementary consequence of a finite signal speed, that events have only a partial ordering and a remote event is unknowable until its signal arrives [13].

This paper adds that light-speed limit to the self-replicating slingshot model and measures what it costs. The answer, built from a controlled paired-ensemble experiment, has two parts, and the first is a caution. The cost is easy to overstate: a simulation that advances in a fixed timestep coarser than the interval between probe launches makes many probes decide from the same stale snapshot at once, and they collide far more than they physically would. Read that way the delay appears to slow the fill of the field by tens of percent. It does not. When the dynamics are resolved so that each probe decides at its own arrival time, the slowdown vanishes: light-speed lag does not measurably delay the fill, and the slingshot speed-up

survives contact with finite information intact, reproducing the source model quantitatively. The real cost shows up elsewhere. Every star still gets settled, and with a negligible change in how many probes are built, well under a percent; what the delay changes is how many wasted journeys the swarm flies to stars other probes have already claimed. That is a cost in fuel and effort, not in time, and a perfect-information model hides it entirely. It is governed cleanly by a single dimensionless number, the probe’s speed in units of the speed of light: negligible for a powered cruise, about a percent of journeys at slingshot speeds, and rising to roughly a fifth at the directed-energy speeds that beamed-propulsion concepts target. The coordination tax is paid in fuel, not time, and it scales with how fast the swarm flies.

II. THE LIGHT-DELAYED BELIEF MODEL

A. The knowledge gate

The change to the model is one gate on knowledge. A settled star is treated as an omnidirectional beacon that begins emitting in the year it is settled. A probe deciding its next target at star f in year Y believes a distant star i is settled only once that beacon’s light has had time to arrive:

$$\text{settled}(i) + \frac{\text{dist}(f, i)}{c} \leq Y. \quad (1)$$

Under the perfect-information assumption of the source model, (1) collapses to “settled at any non-negative year,” and the model reproduces our perfect-information baseline bit-for-bit; the light-speed term is what introduces stale views. When two probes both believe an unsettled star is the best target they both fly to it; the loser arrives to find it taken, its trip wasted, and must re-target from the even more stale picture it holds on arrival. This is availability chosen over consistency: rather than wait for a single agreed map that finite light-speed makes impossible, each probe acts now on its local view [13]. Stars carry velocity magnitudes that feed the slingshot boost but are held fixed in position, following the source model, so $\text{dist}(f, i)$ in (1) is static and the beacon geometry does not shift under the probe. Physical arrival always reads ground truth: what a probe finds on arrival is the star’s real status, whatever it believed when it aimed.

B. The three flight policies, and the speed axis

We carry three named policies, following the slingshot studies we extend [9], [10]. *Powered nearest-neighbour* flight moves under the probe’s own power to the nearest unsettled star. *Nearest-star slingshot* flight takes the same nearest target but gains speed from gravitational slingshots off the stars it passes. *Maximum-boost slingshot* flight instead selects, among the nearest believed-unsettled stars, the target that yields the largest velocity gain. The gravitational boost self-limits (its per-encounter gain falls off once the probe outruns the stellar escape speed), so both slingshot policies saturate at a mean speed of a few thousand km/s, a few times 10^{-3} of c . To reach the higher speeds that beamed directed-energy concepts target, 0.1 to 0.2 c [11], [12], a probe must be driven, not slung;

we therefore sweep the powered cruise speed as a clean, direct control on v/c , from the source model’s $3 \times 10^{-5} c$ up to 0.2 c . Powered flight thus spans the whole speed axis, while the two slingshot policies sit near $v/c \approx 0.01$.

C. Event-driven resolution, and why it is necessary

The fold can advance either by a fixed timestep or by jumping to the next probe arrival. This is not a mere numerical convenience, and getting it wrong is the single largest source of error in this problem. A fixed timestep Δt processes together every probe that arrives within a window, so all of them decide against the same snapshot of the settled map. When Δt exceeds the interval between launches – which it does badly in the boosted regime, where a probe crosses a parsec in centuries against a source-model step of thousands of years – the window batches many independent decisions into one synchronized burst, and they collide far more often than they would if each decided at its true arrival time. The measured coordination cost then reflects the size of the timestep, not the physics. We therefore run the model event-driven throughout: the fold jumps to the earliest outstanding arrival, advances the clock to it, and processes exactly the probes arriving then. This is the exact $\Delta t \rightarrow 0$ limit, dt-independent by construction, and it is what every number in this paper is computed on. Section III shows directly that the apparent fill-time tax is an artifact of a coarse step and collapses to zero as the step resolves.

D. The governing ratio

How much the gate matters is governed by one dimensionless number: the information delay across a hop divided by the travel time across it. For a hop of length d the delay is d/c and the travel time is d/v , so the ratio is

$$\Lambda = \frac{d/c}{d/v} = \frac{v}{c}, \quad (2)$$

the probe’s speed in units of c , independent of the hop length. A slow probe ($\Lambda \ll 1$) outpaces its own ignorance only trivially, so stale views barely arise; a fast probe carries a large Λ and routinely decides before the relevant news can reach it. At the powered cruise speed $\Lambda \approx 3 \times 10^{-5}$ is negligible. As we show below, once the dynamics are resolved Λ does not predict a fill-time penalty (there is essentially none), but it *does* govern the redundant-travel penalty cleanly and monotonically – it is the right parameter, for the right cost.

E. Experimental design and parameters

The field is a box of stars at a uniform density of one star per cubic parsec, the convention of the source model. Each settled star launches two offspring (the replication branching factor); branching is what makes two probes ever contend for a star, so it is a precondition for any coordination effect at all, and the field fills exponentially rather than as a single roving chain. Three properties keep the measurement honest. First, knowledge is evaluated only at the decision star at the moment of decision; news a probe’s path happens to cross in mid-flight

is ignored, which deliberately undercounts what a probe could know and so makes the measured waste a conservative upper bound. Second, a probe that loses a race re-targets, but only up to a fixed number of times before it is retired; this cap is applied identically under both coordination regimes, so the paired comparison is fair, and the results are insensitive to its value. Third, the whole simulation is a pure, seeded, event-driven fold: fixing the seed fixes the star field, every arrival, and every target choice, so a run with the light-speed gate and a run without it can be compared on the *identical* galaxy. That paired design attributes a difference to the delay alone rather than to the luck of the star layout.

Two dependencies are worth stating because they are easy to over-read. Changing the value of the uniform density rescales every inter-star distance by a common factor, multiplying travel time and light-travel time on every hop equally, so Λ and the relative penalties are unchanged (a pure geometric rescaling of the absolute clock); we confirm this holds to the printed digit across densities from the sparse solar neighbourhood [14] near 0.14 up to 5 stars per cubic parsec. And because the field always fills under both regimes and each settlement launches a fixed number of offspring, the two regimes build essentially the same number of probes – they differ only in a handful of terminal launches as targets run out, well under a percent – so the resource the delay actually moves is the number of journeys flown, which is why the coordination cost is a travel cost.

F. Baseline validation

By construction the “instant” regime is the $c \rightarrow \infty$ limit of (1), and it reproduces the plain perfect-information fold bit-for-bit. Against Nicholson and Forgan’s published model [10] the event-driven fold reproduces both headline findings *quantitatively*: gravitational slingshots explore about two orders of magnitude faster than powered flight (a factor of about 166 on a 400-star field), and their surprising result that aiming for the nearest star beats aiming for the largest boost on total exploration time holds, while the maximum-boost policy reaches the higher peak speed. A coarse fixed step had quantized the boosted hops and undercut this speed-up to about twenty-fold; resolving the step recovers the full factor, which is the same correction that removes the spurious fill-time tax.

III. RESULTS: THE COORDINATION TAX

We measured the cost as a paired ensemble: 32 seeded galaxies, each filled twice on the *identical* field, once with perfect information and once with the light-speed gate of (1), all at the resolved event-driven timestep. Two quantities matter: the relative slowdown in the time to fill the field, and the number of wasted journeys, trips to stars a probe finds already settled on arrival. Reporting the distribution across the ensemble follows the seeded Monte Carlo practice of the field [8]; we give the median with a seeded percentile-bootstrap 95% confidence interval and a two-sided sign test over the paired seeds.

A. *There is no fill-time tax; the coarse-timestep one is an artifact*

Light-speed lag does not measurably slow the fill of the field. Fig. 2 shows why the literature intuition says otherwise. Run with the source model’s fixed timestep the fill-100% penalty for nearest-star slingshot flight is a median of 30.3%, positive in all 32 seeds – a large, robust-looking tax. It is an artifact of the step. As the timestep is refined the penalty falls monotonically – 27.6%, 25.0%, 12.8%, 5.3% at 2000, 1000, 500, 250 years – and at the event-driven limit it is a median of 0.0%, no longer distinguishable from zero. The coarse step had batched many launches into one window so they decided from the same stale snapshot and collided; resolving the step lets each probe decide at its own arrival time, and the collisions that cost time disappear. The same refinement restores the slingshot speed-up to its full value: against powered flight the event-driven fold reproduces Nicholson and Forgan’s roughly two-orders-of-magnitude figure (about $166\times$ on a 400-star field), where the coarse step had reported only about $20\times$. The celebrated speed-up survives contact with finite information; the fill-time tax does not exist.

B. *The real cost is redundant travel, governed by $\Lambda = v/c$*

What the delay changes is how far the swarm flies. Because the field always fills and each settlement launches a fixed number of offspring, both coordination regimes build essentially the same number of probes – differing by well under a percent, in a handful of terminal launches – so the resource the delay moves is journeys, and stale views add wasted ones. That redundant-travel tax is governed cleanly by the probe’s speed in units of c . Table I and Fig. 1 give it for powered flight swept across the speed axis: negligible at the cruise speed, a median of 0.7% of journeys at $\Lambda = 0.01$, rising monotonically to 18.4% at $\Lambda = 0.2$. It is not marginal – at every speed from $\Lambda = 0.03$ upward it is positive in all 32 seeds ($p \approx 5 \times 10^{-10}$), and Fig. 3 shows the per-seed distributions sitting cleanly above zero. A small fill-time tax rides alongside it once the swarm is fast, growing from nil at $\Lambda \leq 0.01$ to a few percent at $\Lambda = 0.2$ (Table I), but the redundant travel is the larger and cleaner signal, and unlike the coarse-step time tax it is real: it does not vanish when the dynamics are resolved, because it is not a batching artifact but the physical consequence of deciding on stale views.

This is what Λ was always meant to govern. The ratio is the same on every hop, so it cannot pick out where in the field the cost lands; but it sets, cleanly and monotonically, how much of the swarm’s motion is wasted. It is the right dimensionless group for the fuel tax, and it does not predict a time tax because at resolved dynamics there is essentially none to predict.

C. *The fuel tax is a scale-stable fraction of effort*

The tax is not an artifact of the small field either. Fig. 4 sweeps the system size at $\Lambda = 0.2$. The absolute count of wasted journeys grows with the field, as it must – from a median of about 230 extra trips at 300 stars to about 1,160

TABLE I
COORDINATION TAX OVER THE 32-SEED PAIRED ENSEMBLE AT THE EVENT-DRIVEN TIMESTEP, POWERED FLIGHT, AS A FUNCTION OF $\Lambda = v/c$: THE REDUNDANT-TRAVEL (FUEL) TAX AS A PERCENT OF THE WASTED JOURNEYS UNDER PERFECT INFORMATION, AND THE FILL-TIME TAX, EACH AS A MEDIAN WITH A BOOTSTRAP 95% CONFIDENCE INTERVAL. THE FUEL TAX IS POSITIVE IN ALL 32 SEEDS FROM $\Lambda = 0.03$ UP.

$\Lambda = v/c$	Fuel tax (%)	Fill-time tax (%)
0.01	0.7 [0.6, 1.1]	0.0 [0.0, 0.0]
0.03	3.2 [2.1, 4.0]	0.2 [0.0, 1.4]
0.05	5.8 [4.3, 7.3]	1.3 [0.2, 2.4]
0.10	10.9 [7.8, 12.8]	2.4 [0.9, 4.1]
0.20	18.4 [16.9, 21.7]	3.6 [1.9, 7.3]

TABLE II
WHERE REAL PROPULSION CONCEPTS SIT ON THE Λ AXIS, AND THE FUEL TAX EACH INCURS. THE POWERED CRUISE IS FAR TOO SLOW FOR LAG TO BITE; GRAVITATIONAL SLINGSHOTS SELF-LIMIT NEAR $\Lambda \approx 0.01$; ONLY BEAMED DIRECTED-ENERGY PROPULSION REACHES THE REGIME WHERE THE TAX IS LARGE.

Propulsion regime	$\Lambda = v/c$	Fuel tax (%)
Powered cruise [10]	3×10^{-5}	≈ 0
Gravitational slingshot [10]	≈ 0.01	≈ 1
Directed-energy / light sail [11]	0.1 to 0.2	11 to 18

at 1,200 – but as a fraction of the perfect-information waste the tax is flat, 17.9%, 18.9%, 19.2% at 300, 600, 1,200 stars, positive in all 16 seeds at every size. The redundant-travel cost is thus a roughly size-independent fraction of the swarm’s effort, which is what makes the extrapolation toward galactic scale a supported trend rather than a boundary effect of a small box.

D. Where the cost falls

Table II places the physical propulsion concepts on this axis. The source model’s powered cruise ($3 \times 10^{-5} c$) is far too slow for any race to matter, and pays nothing. The gravitational slingshot that motivated this study reaches only $\Lambda \approx 0.01$ before its boost self-limits, and pays about a percent – real, but small. The tax becomes large only for beamed directed-energy or light-sail propulsion at 0.1 to 0.2 c [11], [12], where it takes a fifth of journeys. Every case still fills the field to completion: this is a property of the model, not a result, since with re-targeting and no loss term the reachable field must close, and delay supplies no term that would leave a permanent unsettled remainder of the kind an “Aurora” steady state requires [12]. The coordination failure shows up as wasted motion, not as an unfilled galaxy.

IV. DISCUSSION

The result refines rather than overturns the slingshot picture, and it does so twice over. Nicholson and Forgan’s speed-up is real, large, and here reproduced quantitatively [10]; finite light-speed does not erode it, and does not measurably delay the fill of the field. What a perfect-information model hides is not lost time but lost motion: probes flying to stars that others

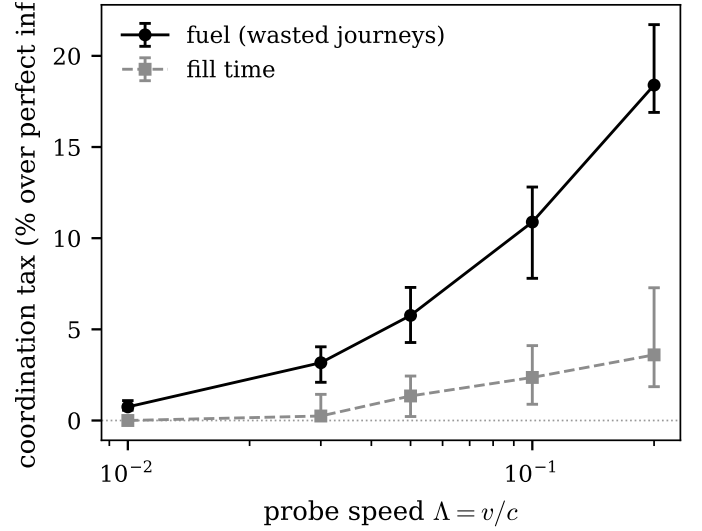


Fig. 1. The coordination tax versus probe speed $\Lambda = v/c$, powered flight, over the 32-seed ensemble at the event-driven timestep (points: medians; bars: bootstrap 95% confidence intervals). The redundant-travel (fuel) tax rises cleanly and monotonically with Λ , from under a percent at slingshot speeds to about a fifth at directed-energy speeds; the fill-time tax is a small companion that only appears once the swarm is fast.

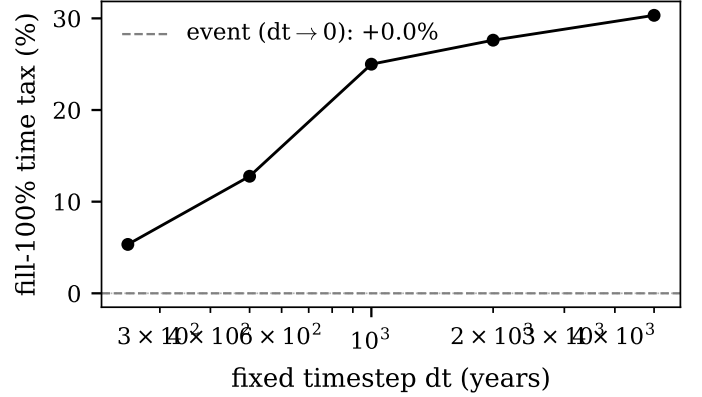


Fig. 2. The fill-100% time penalty for nearest-star slingshot flight versus the fixed timestep, over the 32-seed ensemble. The apparent penalty is monotone in the step and collapses to zero (dashed line: the event-driven, $dt \rightarrow 0$ limit). The tens-of-percent “coordination tax” of a coarse-step model is a discretization artifact, not a physical slowdown.

have already claimed. That redundant travel is the coordination tax, it is governed by the probe’s speed in units of c , and it matters only once a swarm flies fast enough for a decision to outrun its own news.

A. Why the cost is fuel, and larger than it looks

Because the field always fills and each settlement launches a fixed number of offspring, the two coordination regimes build essentially the same number of probes, differing by well under a percent; the delay changes almost nothing about how much is manufactured, only how far it is flown. The tax is therefore a travel cost, and we have measured it as a count of wasted journeys. In energy terms it is larger than that count suggests, and increasingly so with speed: the kinetic cost of a hop scales

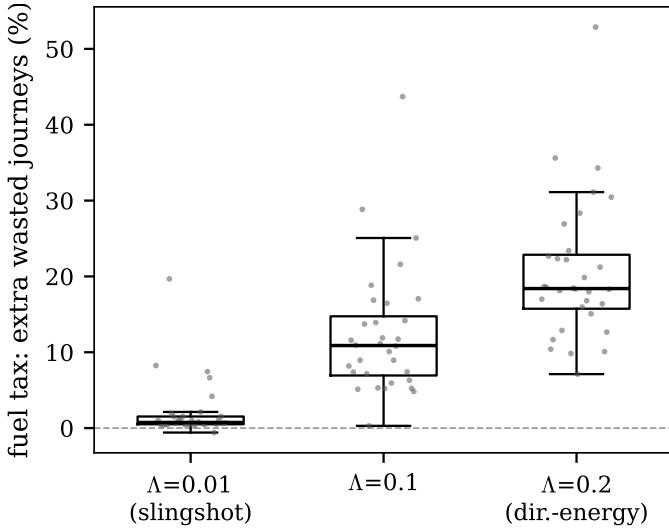


Fig. 3. Per-seed distribution of the fuel tax (extra wasted journeys, as a percent of the perfect-information waste) at slingshot speed ($\Lambda = 0.01$), $\Lambda = 0.1$, and directed-energy speed ($\Lambda = 0.2$), over the 32-seed ensemble (box: interquartile range and median; points: individual seeds). The tax is small but consistently positive at slingshot speed and unambiguous at directed-energy speed.

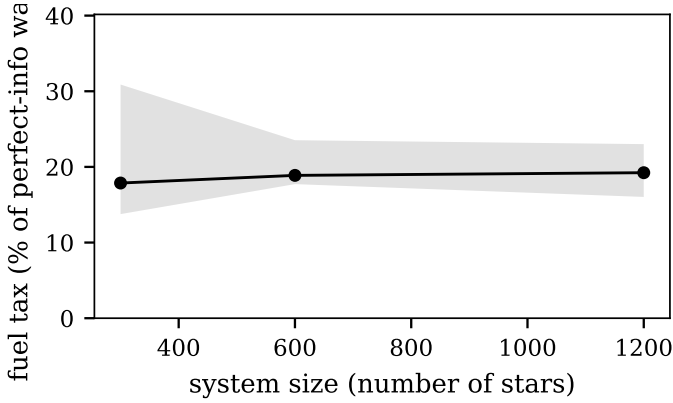


Fig. 4. Fuel tax as a percent of the perfect-information waste versus system size, at $\Lambda = 0.2$ (point: median; shaded band: interquartile range over 16 seeds). The tax is a roughly scale-independent fraction of effort, near 18 to 19% across a four-fold range in the number of stars, so it does not wash out at large scale.

with v^2 , so a wasted journey at $0.2c$ squanders far more than one at slingshot speed, on top of there being proportionally more of them. A coordination tax that is a fifth of journeys at directed-energy speed is a heavier fraction of the energy budget still. This is the sense in which the tax “scales with how fast the swarm flies”: both the rate of wasted trips and the cost of each rise with v .

B. Limitations

The measurement is a conservative upper bound in two senses, and both point to future work. Probes here learn only at their decision stars, ignoring news their paths cross in flight, so a two-endpoint knowledge model would recover some wasted trips; and probes share nothing directly, so probe-

to-probe relay of the settled map would let a swarm partially coordinate rather than merely collide. Both would lower the tax without erasing it, since the floor – that no probe can know a distant star is taken until its light arrives – is set by physics, not protocol. Three further bounds on scope. First, the field is uniform; changing its density only rescales the clock (confirmed), but a clumpier distribution would lengthen the long-range hops and is a genuine unknown we do not probe. Second, at high v/c the hops are short, so an event-driven run generates of order one arrival per star settled many times over and costs roughly $O(N^2)$; this bounded the size sweep at a few thousand stars, over which the fuel tax is a stable fraction, so the extrapolation to a full galaxy is a supported trend rather than a measurement. Third, the dynamics have no loss term: settled sites never fail, so the field always closes. A settlement-death term of the kind that produces the “Aurora” steady state [12] is what would be needed to turn coordination failure into a permanent unsettled remainder; delay alone does not.

C. The signature of delayed agreement

The behaviour is, by analogy rather than derivation, the expected signature of delayed multi-agent agreement. Formal consensus degrades as the ratio of communication delay to decision cadence grows: the standard protocol is stable only while the one-hop delay stays below a bound set by the coupling [15], generalized later to switching topologies and link failures [16]. We do not apply that bound quantitatively – it needs a probe decision cadence the literature does not fix – but the regime is unambiguous: at a parsec spacing the round-trip light-time is years, far longer than any plausible cadence, so the probes sit deep in the region where guaranteed agreement is not merely slow but impossible [17], and a system that must keep acting trades a single consistent map for availability [18], which is exactly what (1) encodes. The same progression drove the space teleoperation and networking communities from closed-loop control through move-and-wait and supervisory control [19], [20] to hop-by-hop delay-tolerant store-and-forward once no continuous path survives [21]–[23]. At interstellar hops every faster regime is left behind; each probe is an independent colony acting on priors, and the coordination tax is what that independence costs in fuel when probes want the same stars.

D. A design lesson

The corollary is a threshold, not a preference. At slingshot speeds the tax is about a percent of journeys: coordination machinery – relay, gossip, a shared map – is not worth its mass, and a designer should spend everything on speed. At the directed-energy speeds that make a probe fast enough to matter, the same physics that buys the speed also makes stale views expensive, and a fifth of journeys wasted is worth engineering against. The faster the swarm you can build, the more a little coordination is worth buying – and the finite speed of light sets how much you can ever buy back.

V. CONCLUSION

Adding a finite speed of information to the self-replicating slingshot model turns a convenient assumption into a measurable cost, but not the cost a coarse model reports. Once the dynamics are resolved so that each probe decides at its true arrival time, light-speed lag does not measurably slow the fill of the field, and the slingshot speed-up survives intact, reproducing Nicholson and Forgan quantitatively; the tens-of-percent fill-time penalty seen in a fixed-timestep model is a discretization artifact. The genuine cost is redundant travel. Every star is still settled, with a negligible change in the number of probes built, but stale views send probes to stars other probes have already claimed, and that waste is governed cleanly by the probe's speed in units of the speed of light, $\Lambda = v/c$: negligible for a powered cruise, about a percent of journeys at slingshot speeds, and roughly a fifth at the directed-energy speeds beamed propulsion targets, heavier still as a fraction of the energy budget because a fast wasted hop costs more. The coordination tax a perfect-information model hides is paid in fuel, not time, and it scales with how fast the swarm flies. Two extensions – knowledge gained in flight, and probe-to-probe relay of the settled map – would lower it; neither can remove the physical floor beneath it, and how much they can buy back is itself bounded by the finite speed of light.

DATA AVAILABILITY

The model is a deterministic, seeded, event-driven fold; every figure, table, and reported statistic regenerates from source by running the swarm module's experiment and figure scripts, and every quantitative claim traces to a cited source in the project's shared bibliography. The code and its citation metadata are archived with the concept DOI 10.5281/zenodo.21249296.

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